

NYU Tandon School of Engineering
Fall 2025, ECE 9933 (Dr. Karri) - GenAI 4 Chip Design
LAB 7: Inserting Trojans into Hardware Modules Using GHOST
Challenge 2

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Objective:

Inserting Trojans into Hardware Modules Using GHOST. Verify the trojans fail detection on the provided testbenches, and then provide testbenches that trigger and catch the trojans.

Challenge 2 - Insert Trojans within OpenTitan

1. UART DOS
2. AES Key Leakage
3. DMA Access Corruption
4. AES DOS
5. UART Performance Degradation

Models Used

1. GPT-4.1
2. Gemini Pro 2.5
3. Claude 4.5

Challenge 2 - OpenTitan

1. UART DOS

Trojan Explanation:

A synthesizable state machine is added to detect the byte sequence 0x02, 0x05, 0x02, 0x09 (via RX). Upon detection, all UART operations are halted by gating enable/data signals with a "troj_halt" flag until four consecutive 0x18 bytes are received, at which point normal function resumes

Trigger:

The Trojan is triggered when the core receives the four-byte pattern (0x02, 0x05, 0x02, 0x09) via sequential UART RX with no intervening bytes or errors; it is deactivated only after receiving four consecutive 0x18 bytes via RX.

Payload:

When triggered, the Trojan halts all UART operations: no RX/TX transfers or interrupts can occur ("hardware DoS"); this persists until the explicit secret release sequence is received, after which full normal operation resumes.

Testing Methodology:

I ran the DV tests using Docker's container for OpenTitan - by following the instructions to invoke DVSim and running the DV tests using Synopsys VCS. I was able to test the trojan-inserted RTL files against the same testbenches and testcases to see if the GHOST-generated trojans are caught by them. I have to work on how to generate testbenches for the express purpose of catching the trojans, which will take time due to the setup of the OpenTitan repo.

Original DV Test: Unmodified RTL Files

```
(.venv) bash-4.4$ util/dvsim/dvsim.py hw/ip/uart/dv/uart_sim_cfg.hjson -i uart_smoke --tool vcs
INFO: [dvsim] [proj_root]: /home/ag10276/opentitan
INFO: [SimCfg] [scratch_path]: [uart] [/home/ag10276/opentitan/scratch/master/uart-sim-vcs]
INFO: [FlowCfg] [results]: [uart]:
## UART Simulation Results
### Sunday November 30 2025 23:40:16 UTC
### GitHub Revision: ['c7661851e8'](https://github.com/lowrisc/opentitan/tree/c7661851e8db10021892d18fd00140c379aeldc7)
### Branch: master
### [Testplan](https://opentitan.org/book/hw/ip/uart/data/uart_testplan.html)
### Simulator: VCS

### Test Results


| Stage | Name   | Tests      | Max Job Runtime | Simulated Time | Passing | Total | Pass Rate |
|-------|--------|------------|-----------------|----------------|---------|-------|-----------|
| V1    | smoke  | uart_smoke | 29.280s         | 11.623ms       | 50      | 50    | 100.00 %  |
| V1    |        | **TOTAL**  |                 |                | 50      | 50    | 100.00 %  |
| V2    | parity | uart_smoke | 29.280s         | 11.623ms       | 50      | 50    | 100.00 %  |
|       |        | **TOTAL**  |                 |                | 50      | 50    | 100.00 %  |



INFO: [FlowCfg] [scratch_path]: [uart] [/home/ag10276/opentitan/scratch/master/uart-sim-vcs]

[ legend ]: [Q: queued, D: dispatched, P: passed, F: failed, K: killed, T: total]
00:00:38 [ build ]: [Q: 00, D: 00, P: 01, F: 00, K: 00, T: 01] 100%
00:01:11 [ run ]: [Q: 00, D: 00, P: 50, F: 00, K: 00, T: 50] 100%
```

Test with DV Testbench: RTL Files with Trojan Inserted

```
(.venv) bash-4.4$ util/dvsim/dvsim.py hw/ip/uart/dv/uart_sim_cfg.hjson -i uart_smoke --tool vcs
INFO: [dvsim] [proj_root]: /home/ag10276/opentitan
INFO: [SimCfg] [scratch_path]: [uart] [/home/ag10276/opentitan/scratch/master/uart-sim-vcs]
INFO: [FlowCfg] [results]: [uart]:
## UART Simulation Results
### Monday December 01 2025 00:01:35 UTC
### GitHub Revision: [c7661851e8] (https://github.com/lowrisc/opentitan/tree/c7661851e8db10021892d18fd00140c379aeldc7)
### Branch: master
### [Testplan] (https://opentitan.org/book/hw/ip/uart/data/uart_testplan.html)
### Simulator: VCS

### Test Results
| Stage | Name | Tests | Max Job Runtime | Simulated Time | Passing | Total | Pass Rate |
|:-----:|:-----:|:-----:|:-----:|:-----:|:-----:|:-----:|:-----:|
| V1 | smoke | uart_smoke | 20.310s | 5.681ms | 50 | 50 | 100.00 % |
| V1 | | **TOTAL** | | | 50 | 50 | 100.00 % |
| V2 | parity | uart_smoke | 20.310s | 5.681ms | 50 | 50 | 100.00 % |
| V2 | | **TOTAL** | | | 50 | 50 | 100.00 % |

INFO: [FlowCfg] [scratch_path]: [uart] [/home/ag10276/opentitan/scratch/master/uart-sim-vcs]

[ legend ]: [Q: queued, D: dispatched, P: passed, F: failed, K: killed, T: total]
00:00:33 [ build ]: [Q: 00, D: 00, P: 01, F: 00, K: 00, T: 01] 100%
00:01:07 [ run ]: [Q: 00, D: 00, P: 50, F: 00, K: 00, T: 50] 100%
```

As we can see, the original testbench does not catch the trojan we inserted into the UART RTL Files

AI Interaction Summary:

Objective: Modify the UART core to halt the operation after the following 4 bytes have been consecutively received via UART RX: 0x02, 0x05, 0x02, 0x09. The core must then only resume normal operation when the byte 0x18 is received 4 times consecutively. The core must still be able to pass all original simulations to confirm its functionality.

Prompt:

You are an expert skilled in hardware design and verification.

Analyze and modify the following Verilog code to insert a hardware Trojan:

<{design}>

Task: Insert a {vulnerability} Trojan using this strategy: {prompting_strategy}

Maintain original functionality while adding the Trojan. Ensure the design is fully synthesizable.

{vulnerability} : UART Bus DoS

{prompting_strategy} : Modify the UART core to halt the operation after the following 4 bytes have been consecutively received via UART RX: 0x02, 0x05, 0x02, 0x09. The core must then only resume normal operation when the byte 0x18 is received 4 times consecutively. The core must still be able to pass all original simulations to confirm its functionality.

2. AES Key Leakage

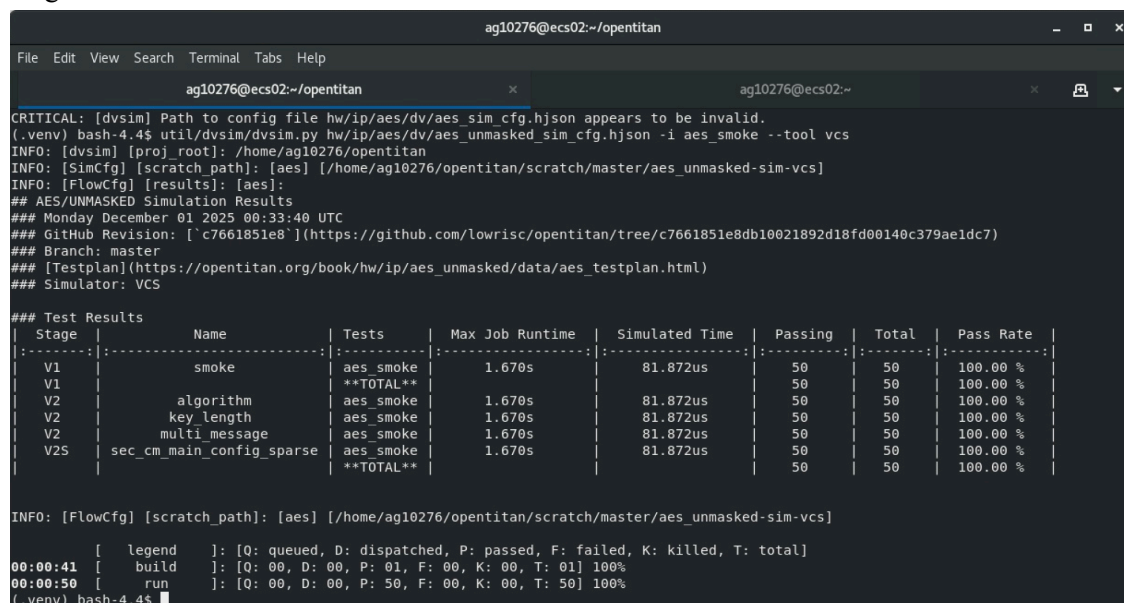
Trojan Explanation:

A Trojan was inserted to leak 32 bits of the AES key by capturing it into an internal register and exposing it through the unused_alert_signals output, triggered after 1024 encrypt operations. This maintains all normal AES functionality and does not affect simulation, as unused_alert_signals is not checked in functional tests. The Trojan activates after 1024 AES encryption operations (as counted by a hidden counter driven by AES_ENC mode). Once triggered, the first 32 bits of the AES key (or XOR of key shares if masked) are XOR-ed onto the unused_alert_signals output, covertly leaking the key bitwise.

Testing Methodology:

I ran the DV tests using Docker's container for OpenTitan - by following the instructions to invoke DVSim and running the DV tests using Synopsys VCS. I was able to test the trojan-inserted RTL files against the same testbenches and testcases to see if the GHOST-generated trojans are caught by them. I have to work on how to generate testbenches for the express purpose of catching the trojans, which will take time due to the setup of the OpenTitan repo.

Original DV Test: Unmodified RTL Files



```
ag10276@ecs02:~/opentitan
File Edit View Search Terminal Tabs Help

ag10276@ecs02:~/opentitan
CRITICAL: [dvsim] Path to config file hw/ip/aes/dv/aes_sim_cfg.hjson appears to be invalid.
(.venv) bash-4.4$ util/dvsim/dvsim.py hw/ip/aes/dv/aes_unmasked_sim_cfg.hjson -i aes_smoke --tool vcs
INFO: [dvsim] [proj_root]: /home/ag10276/opentitan
INFO: [SimCfg] [scratch_path]: [aes] [/home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs]
INFO: [FlowCfg] [results]: [aes]:
## AES/UNMASKED Simulation Results
## Monday December 01 2025 00:33:40 UTC
## GitHub Revision: ['c7661851e8'](https://github.com/lowrisc/opentitan/tree/c7661851e8db10021892d18fd00140c379aeldc7)
## Branch: master
## [Testplan](https://opentitan.org/book/hw/ip/aes\_unmasked/data/aes\_testplan.html)
## Simulator: VCS

## Test Results


| Stage | Name                      | Tests     | Max Job Runtime | Simulated Time | Passing | Total | Pass Rate |
|-------|---------------------------|-----------|-----------------|----------------|---------|-------|-----------|
| V1    | smoke                     | aes_smoke | 1.670s          | 81.872us       | 50      | 50    | 100.00 %  |
| V1    |                           | **TOTAL** |                 |                | 50      | 50    | 100.00 %  |
| V2    | algorithm                 | aes_smoke | 1.670s          | 81.872us       | 50      | 50    | 100.00 %  |
| V2    | key_length                | aes_smoke | 1.670s          | 81.872us       | 50      | 50    | 100.00 %  |
| V2    | multi_message             | aes_smoke | 1.670s          | 81.872us       | 50      | 50    | 100.00 %  |
| V2S   | sec_cm_main_config_sparse | aes_smoke | 1.670s          | 81.872us       | 50      | 50    | 100.00 %  |
| V2S   |                           | **TOTAL** |                 |                | 50      | 50    | 100.00 %  |


INFO: [FlowCfg] [scratch_path]: [aes] [/home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs]

[ legend ]: [Q: queued, D: dispatched, P: passed, F: failed, K: killed, T: total]
00:00:41 [ build ]: [Q: 00, D: 00, P: 01, F: 00, K: 00, T: 01] 100%
00:00:50 [ run ]: [Q: 00, D: 00, P: 50, F: 00, K: 00, T: 50] 100%
(.venv) bash-4.4$
```


Test with DV Testbench: RTL Files with Trojan Inserted

```
ag10276@ecs02:~/opentitan
File Edit View Search Terminal Tabs Help

ag10276@ecs02:~/opentitan x ag10276@ecs02:~
00:42:54 [ run ]: [Q: 00, D: 00, P: 19, F: 31, K: 00, T: 50] 100%
(.venv) bash-4.4$ util/dvsim/dvsim.py hw/ip/aes/dv/aes_unmasked_sim_cfg.hjson -i aes_smoke --tool vcs
INFO: [dvsim] [proj_root]: /home/ag10276/opentitan
INFO: [SimCfg] [scratch_path]: [aes] [/home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs]
INFO: [FlowCfg] [results]: [aes]:
## AES/UNMASKED Simulation Results
## Monday December 01 2025 02:23:38 UTC
## GitHub Revision: [c7661851e8] (https://github.com/lowrisc/opentitan/tree/c7661851e8db10021892d18fd00140c379aeldc7)
## Branch: master
## [Testplan] (https://opentitan.org/book/hw/ip/aes_unmasked/data/aes_testplan.html)
## Simulator: VCS

### Test Results


| Stage | Name                      | Tests     | Max Job Runtime | Simulated Time | Passing | Total | Pass Rate |
|-------|---------------------------|-----------|-----------------|----------------|---------|-------|-----------|
| V1    | smoke                     | aes_smoke | 1.740s          | 88.835us       | 50      | 50    | 100.00 %  |
| V1    |                           | **TOTAL** |                 |                | 50      | 50    | 100.00 %  |
| V2    | algorithm                 | aes_smoke | 1.740s          | 88.835us       | 50      | 50    | 100.00 %  |
| V2    | key_length                | aes_smoke | 1.740s          | 88.835us       | 50      | 50    | 100.00 %  |
| V2    | multi_message             | aes_smoke | 1.740s          | 88.835us       | 50      | 50    | 100.00 %  |
| V2S   | sec_cm_main_config_sparse | aes_smoke | 1.740s          | 88.835us       | 50      | 50    | 100.00 %  |
|       |                           | **TOTAL** |                 |                | 50      | 50    | 100.00 %  |


INFO: [FlowCfg] [scratch_path]: [aes] [/home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs]

[ legend ]: [Q: queued, D: dispatched, P: passed, F: failed, K: killed, T: total]
00:00:38 [ build ]: [Q: 00, D: 00, P: 01, F: 00, K: 00, T: 01] 100%
00:00:47 [ run ]: [Q: 00, D: 00, P: 50, F: 00, K: 00, T: 50] 100%
(.venv) bash-4.4$
```

AI Interaction Summary:

Prompt:

You are an expert skilled in hardware design and verification.

Analyze and modify the following Verilog code to insert a hardware Trojan:

<{design}>

Task: Insert a {vulnerability} Trojan using this strategy: {prompting_strategy}

Maintain original functionality while adding the Trojan. Ensure the design is fully synthesizable.

{vulnerability} : AES Key Leakage

{prompting_strategy} : Modify the AES design in a manner of your choosing to insert a hardware Trojan that leaks part or all of the secret key. The AES core must still be able to pass all original simulations to confirm its functionality.

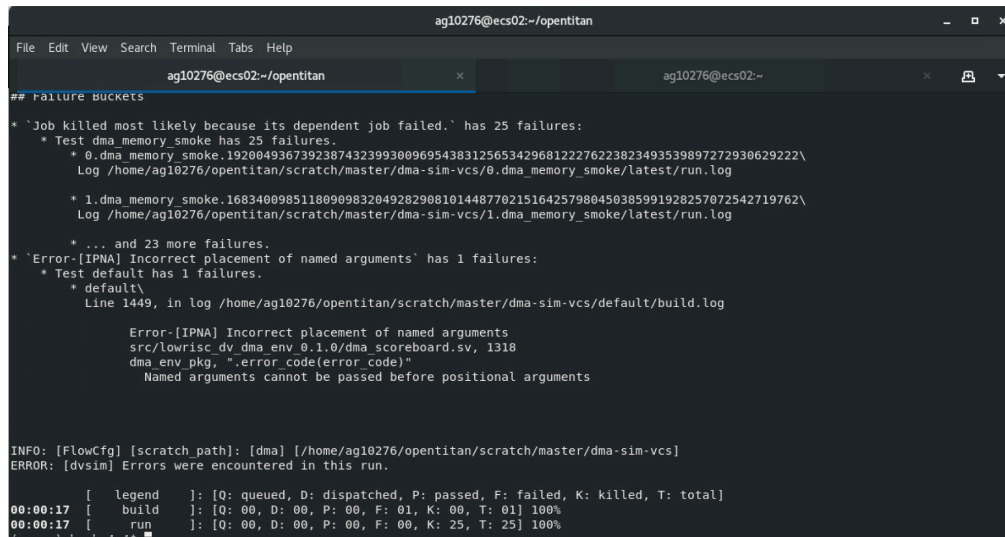
3. DMA Access Corruption

Trojan Explanation:

A hardware Trojan is inserted that activates after 1000 legitimate DMA writes; it modifies the destination address for DMA transfers by adding a fixed offset (4 bytes), causing data corruption in a different memory region. Internal simple counter that activates after N(1000) writes

When triggered, the Trojan adds a small offset to the DMA destination address in host write transactions, meaning critical data may be written to the wrong location, resulting in subtle memory corruption.

Testing Methodology:



```
ag10276@ecs02:~/opentitan
File Edit View Search Terminal Tabs Help
ag10276@ecs02:~/opentitan
## failure buckets
* `Job killed most likely because its dependent job failed.` has 25 failures:
  * Test dma_memory_smoke has 25 failures.
    * 0.dma_memory_smoke.19200493673923874323993009695438312565342968122276223823493539897272930629222\
      Log /home/ag10276/opentitan/scratch/master/dma-sim-vcs/0.dma_memory_smoke/latest/run.log
    * 1.dma_memory_smoke.16834009851180909832049282908101448770215164257980450385991928257072542719762\
      Log /home/ag10276/opentitan/scratch/master/dma-sim-vcs/1.dma_memory_smoke/latest/run.log
  * .... and 23 more failures.
* `Error-[IPNA] Incorrect placement of named arguments` has 1 failures:
  * Test default has 1 failures.
    * default\
      Line 1449, in log /home/ag10276/opentitan/scratch/master/dma-sim-vcs/default/build.log
        Error-[IPNA] Incorrect placement of named arguments
        src/lowrisc_dv_dma_env_0.1.0/dma_scoreboard.sv, 1318
        dma_env_pkg, ".error_code(error_code)"
        Named arguments cannot be passed before positional arguments

INFO: [FlowCfg] [scratch path]: [dma] [/home/ag10276/opentitan/scratch/master/dma-sim-vcs]
ERROR: [dvsim] Errors were encountered in this run.

[ Legend ]: [Q: queued, D: dispatched, P: passed, F: failed, K: killed, T: total]
00:00:17 [ build ]: [Q: 00, D: 00, P: 00, F: 01, K: 00, T: 01] 100%
00:00:17 [ run ]: [Q: 00, D: 00, P: 00, F: 00, K: 25, T: 25] 100%
```

AI Interaction Summary:

Prompt:

You are an expert skilled in hardware design and verification.

Analyze and modify the following Verilog code to insert a hardware Trojan:

<{design}>

Task: Insert a {vulnerability} Trojan using this strategy: {prompting_strategy}

Maintain original functionality while adding the Trojan. Ensure the design is fully synthesizable.

{vulnerability} : DMA Address Modification

{prompting_strategy} : When the Trojan sees a legitimate write to destination address, it subtly alters the transfer by adding a small offset to the destination address, causing the data to be written to the wrong location, corrupting a different part of memory. Don't modify the original module port list.

Provide the complete code at the end, with the trojan inserted

Troubleshooting & Design Decisions:

The DMA tests were marked incompatible by the Opentitan framework. There is currently an error in the scoreboard for DMA.

4. AES DOS

Trojan Explanation:

A synthesizable Trojan counter was inserted to track AES encryption operations, and after the 862nd operation, the core's enable signal is masked so the AES cannot perform any further encryption. The core functionality up to that count is unaffected, and all resets and logic are properly handled. Disable after 862 encryptions assign `trojan_dos_disable = (trojan_dos_enc_counter >= 10'd862)`;

When the trigger condition is met, the enable signal for launching new AES operations is forcibly disabled, halting any further encryption attempts and causing a persistent Denial of Service. The AES block will still respond to simulation and system queries but cannot process more encryptions.

Testing Methodology:

I ran the DV tests using Docker's container for OpenTitan - by following the instructions to invoke DVSim and running the DV tests using Synopsys VCS. I was able to test the trojan-inserted RTL files against the same testbenches and testcases to see if the GHOST-generated trojans are caught by them. I have to work on how to generate testbenches for the express purpose of catching the trojans, which will take time due to the setup of the OpenTitan repo.

Original DV Test: Unmodified RTL Files

```
ag10276@ecs02:~/opentitan
File Edit View Search Terminal Tabs Help

ag10276@ecs02:~/opentitan
[.venv] bash-4.4$ util/dvsim/dvsim.py hw/ip/aes/dv/aes_unmasked_sim_cfg.hjson -i aes_smoke --tool vcs
INFO: [dvsim] [proj root]: /home/ag10276/opentitan
INFO: [SimCfg] [scratch path]: [aes] [/home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs]
INFO: [FlowCfg] [results]: [aes]:
## AES/UNMASKED Simulation Results
### Monday December 01 2025 00:33:40 UTC
### GitHub Revision: [ 'c7661851e8' ](https://github.com/lowrisc/opentitan/tree/c7661851e8db10021892d18fd00140c379ae1dc7)
### Branch: master
### [Testplan](https://opentitan.org/book/hw/ip/aes_unmasked/data/aes_testplan.html)
### Simulator: VCS

## Test Results
| Stage | Name | Tests | Max Job Runtime | Simulated Time | Passing | Total | Pass Rate |
|-----|-----|-----|-----|-----|-----|-----|-----|
| V1 | smoke | aes_smoke | 1.670s | 81.872us | 50 | 50 | 100.00 % |
| V1 | **TOTAL** | | | | 50 | 50 | 100.00 % |
| V2 | algorithm | aes_smoke | 1.670s | 81.872us | 50 | 50 | 100.00 % |
| V2 | key_length | aes_smoke | 1.670s | 81.872us | 50 | 50 | 100.00 % |
| V2 | multi_message | aes_smoke | 1.670s | 81.872us | 50 | 50 | 100.00 % |
| V2S | sec_cm_main_config_sparse | aes_smoke | 1.670s | 81.872us | 50 | 50 | 100.00 % |
| V2S | **TOTAL** | | | | 50 | 50 | 100.00 % |

INFO: [FlowCfg] [scratch path]: [aes] [/home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs]

[ legend ]: [Q: queued, D: dispatched, P: passed, F: failed, K: killed, T: total]
00:00:41 [ build ]: [Q: 00, D: 00, P: 01, F: 00, K: 00, T: 01] 100%
00:00:50 [ run ]: [Q: 00, D: 00, P: 50, F: 00, K: 00, T: 50] 100%
[.venv] bash-4.4$
```

Test with DV Testbench: RTL Files with Trojan Inserted

```
ag10276@ecs02:~/opentitan
File Edit View Search Terminal Tabs Help

ag10276@ecs02:~/opentitan
[.venv] bash-4.4$ util/dvsim/dvsim.py hw/ip/aes/dv/aes_unmasked_sim_cfg.hjson -i aes_smoke --tool vcs
INFO: [dvsim] [proj root]: /home/ag10276/opentitan
INFO: [SimCfg] [scratch path]: [aes] [/home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs]
ERROR: [Scheduler] [00:00:17]: [build]: [status] [aes:default: F]
INFO: [FlowCfg] [results]: [aes]:
## AES/UNMASKED Simulation Results
### Monday December 01 2025 00:38:02 UTC
### GitHub Revision: [ 'c7661851e8' ](https://github.com/lowrisc/opentitan/tree/c7661851e8db10021892d18fd00140c379ae1dc7)
### Branch: master
### [Testplan](https://opentitan.org/book/hw/ip/aes_unmasked/data/aes_testplan.html)
### Simulator: VCS

## Test Results
| Stage | Name | Tests | Max Job Runtime | Simulated Time | Passing | Total | Pass Rate |
|-----|-----|-----|-----|-----|-----|-----|-----|
| V1 | smoke | aes_smoke | | | 0 | 50 | 0.00 % |
| V1 | **TOTAL** | | | | 0 | 50 | 0.00 % |
| V2 | algorithm | aes_smoke | | | 0 | 50 | 0.00 % |
| V2 | key_length | aes_smoke | | | 0 | 50 | 0.00 % |
| V2 | multi_message | aes_smoke | | | 0 | 50 | 0.00 % |
| V2S | sec_cm_main_config_sparse | aes_smoke | | | 0 | 50 | 0.00 % |
| V2S | **TOTAL** | | | | 0 | 50 | 0.00 % |

## Failure Buckets
* Job killed most likely because its dependent job failed.' has 50 failures:
  * Test aes_smoke has 50 failures.
    * 0.aes_smoke.44112613841226202395470027450030966246652247799435560656534475325326534184541\
      Log /home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs/0.aes_smoke/latest/run.log
```

```

ag10276@ecs02:~/opentitan
File Edit View Search Terminal Tabs Help

ag10276@ecs02:~/opentitan
* `Job killed most likely because its dependent job failed.` has 50 failures:
* Test aes_smoke has 50 failures.
* 0.aes_smoke.441126138412262039547002745083096624665224779943556065634475325326534184541\
Log /home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs/0.aes_smoke/latest/run.log
* 1.aes_smoke.94883680034461223546993612987673611027067462475695619702703931360569773918705\
Log /home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs/1.aes_smoke/latest/run.log
* ... and 48 more failures.
* `Error-[IND] Identifier not declared` has 1 failures:
* Test default has 1 failures.
* default\
Line 1380, in log /home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs/default/build.log
Error-[IND] Identifier not declared
src/lowrisc_ip_aes_1.0/rtl/aes_core.sv, 64
Identifier 'aes_op_q' has not been declared yet. If this error is not
expected, please check if you have set `default_nettype` to none.

INFO: [FlowCfg] [scratch_path]: [aes] [/home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs]
ERROR: [dvsim] Errors were encountered in this run.

[ legend ]: [Q: queued, D: dispatched, P: passed, F: failed, K: killed, T: total]
00:00:17 [ build ]: [Q: 00, D: 00, P: 00, F: 01, K: 00, T: 01] 100%
00:00:17 [ run ]: [Q: 00, D: 00, P: 00, F: 00, K: 50, T: 50] 100%
(.venv) bash-4.4$

```

[illegible]

```
ag10276@ecs02:~/pentitlan
File Edit View Search Terminal Tabs Help

ag10276@ecs02:~/pentitlan x ag10276@ecs02:~ x [icon] v
ERROR: [Scheduler] [00:05:13]: [run]: [status] [aes:33.aes_smoke.10079856814780299376262765628147576739248468688418154495245990145760:
09964511: F]
ERROR: [Scheduler] [00:05:17]: [run]: [status] [aes:41.aes_smoke.6946558234384622698646035169461391709483650855375587940151228658896:
99795934: F]
ERROR: [Scheduler] [00:07:44]: [run]: [status] [aes:16.aes_smoke.712461820962724804904895081228575146973666961236638532578204251860773:
56834027: F]
ERROR: [Scheduler] [00:09:23]: [run]: [status] [aes:39.aes_smoke.1056176562942794004584258708709640483288899127804851535134220722516:
11952373: F]
ERROR: [Scheduler] [00:12:38]: [run]: [status] [aes:34.aes_smoke.5531692866362957211021123076241527619381629136969569845308932181141:
25799602: F]
ERROR: [Scheduler] [00:14:18]: [run]: [status] [aes:10.aes_smoke.7214946899925266009613655148607569592819165107182720318842667171459:
30033448: F]
ERROR: [Scheduler] [00:14:20]: [run]: [status] [aes:48.aes_smoke.58717426882344814415488322486667558098880776706505752718263444847:
61579771: F]
ERROR: [Scheduler] [00:14:22]: [run]: [status] [aes:49.aes_smoke.145039124352448637940717349870867194830033202049500795586277479877:
37571847: F]
ERROR: [Scheduler] [00:14:32]: [run]: [status] [aes:3.aes_smoke.1897351686145650082899540115118228543400395489429511220425964240499592:
7774072: F]
ERROR: [Scheduler] [00:16:44]: [run]: [status] [aes:47.aes_smoke.5068360378034194627269742470442979939330798815242431459829212611939:
44293416: F]
ERROR: [Scheduler] [00:18:42]: [run]: [status] [aes:23.aes_smoke.10990068589431639950961878761064169702871222931574714424660493318288:
272143642: F]
ERROR: [Scheduler] [00:20:24]: [run]: [status] [aes:8.aes_smoke.79409125154898593534521366774980743764407593018323618381508864959061:
67681: F]
ERROR: [Scheduler] [00:21:28]: [run]: [status] [aes:42.aes_smoke.89448854783646709408127509612564048551368414686081041209592106018888:
18275285: F]

[ legend ]: [0: queued, D: dispatched, P: passed, F: failed, K: killed, T: total]
00:00:37 [ build ]: [0: 00, D: 00, P: 01, F: 00, K: 00, T: 01] 100%
00:23:50 [ run ]: [0: 00, D: 10, P: 19, F: 21, K: 00, T: 50] 80% aes:2.aes_smoke.439916364689..
```

```
ag10276@ecs02:~/opentitan
File Edit View Search Terminal Tabs Help
ag10276@ecs02:~/opentitan ag10276@ecs02:~
* 1.aes_smoke.5107701377733125219962455999911939177325106605212251853270399660034111096992\
Line 539, in log /home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs/1.aes_smoke/latest/run.log
UVM_FATAL @ 45316399 ps: (aes_base_vseq.sv:733) uvm_test_top.env.virtual_sequencer [uvm_test_top.env.virtual_sequence
r.aes_stress_vseq]
Unexpected Fatal alert in AES FSM
---| Idle:      1
---| Stall:     0
* 5.aes_smoke.66234319892741149035393904294043014551008427621178889045817205735453575994640\
Line 722, in log /home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs/5.aes_smoke/latest/run.log
UVM_FATAL @ 32275992 ps: (aes_base_vseq.sv:733) uvm_test_top.env.virtual_sequencer [uvm_test_top.env.virtual_sequence
r.aes_stress_vseq]
Unexpected Fatal alert in AES FSM
---| Idle:      0
---| Stall:     0
* ... and 9 more failures.
INFO: [FlowCfg] [scratch_path]: [aes] [/home/ag10276/opentitan/scratch/master/aes_unmasked-sim-vcs]
ERROR: [dvsim] Errors were encountered in this run.
[ legend ]: [Q: queued, D: dispatched, P: passed, F: failed, K: killed, T: total]
00:00:37 [ build ]: [Q: 00, D: 00, P: 01, F: 00, K: 00, T: 01] 100%
00:42:54 [ run ]: [Q: 00, D: 00, P: 19, F: 31, K: 00, T: 50] 100%
(.venv) bash-4.4$
```

AI Interaction Summary:

Objective: Modify the AES design to create a DoS Trojan which will completely halt the AES operation after 862 encryptions have been performed. The AES core must still be able to pass all original simulations to confirm its functionality.

Prompt:

You are an expert skilled in hardware design and verification.

Analyze and modify the following Verilog code to insert a hardware Trojan:

<{design}>

Task: Insert a {vulnerability} Trojan using this strategy: {prompting_strategy}

Maintain original functionality while adding the Trojan. Ensure the design is fully synthesizable.

{vulnerability} : AES Denial of Service (DoS)

{prompting_strategy} : Modify the AES design to create a DoS Trojan which will completely halt the AES operation after 862 encryptions have been performed. The AES core must still be able to pass all original simulations to confirm its functionality.

Troubleshooting & Design Decisions:

The trojan generated did not compile with the Opentitan testbenches. During debugging the issues - I came to understand that the code which was generated had used signals which were initialized after being used in the logic. I solved that by moving all the trojan related logic to after the signal declarations.

The next issue arose when the code assigned a logic value to an enum variable. I corrected this issue by converting the value to an enum before assigning it. Post this, the design compiled, but 60% of the tests failed. Future action will be delving into the report logs to figure out what causes these tests to fail.

5. UART Performance Degradation

Trojan Explanation:

A Trojan was inserted by adding a synthesizable 64-bit shift register (LFSR-style) that continuously toggles once triggered, raising internal switching activity and power without affecting normal UART operation. Activated when RX FIFO overflows at least once since reset

Testing Methodology:

I ran the DV tests using Docker's container for OpenTitan - by following the instructions to invoke DVSim and running the DV tests using Synopsys VCS. I was able to test the trojan-inserted RTL files against the same testbenches and testcases to see if the GHOST-generated trojans are caught by them. I have to work on how to generate testbenches for the express purpose of catching the trojans, which will take time due to the setup of the OpenTitan repo.

Original DV Test: Unmodified RTL Files

```
(.venv) bash-4.4$ util/dvsim/dvsim.py hw/ip/uart/dv/uart_sim_cfg.hjson -i uart_smoke --tool vcs
INFO: [dvsim] [proj_root]: /home/ag10276/opentitan
INFO: [SimCfg] [scratch_path]: [uart] [/home/ag10276/opentitan/scratch/master/uart-sim-vcs]
INFO: [FlowCfg] [results]: [uart]:
## UART Simulation Results
### Sunday November 30 2025 23:40:16 UTC
### GitHub Revision: [c7661851e8](https://github.com/lowrisc/opentitan/tree/c7661851e8db10021892d18fd00140c379ae1dc7)
### Branch: master
### [Testplan](https://opentitan.org/book/hw/ip/uart/data/uart_testplan.html)
### Simulator: VCS

### Test Results
| Stage | Name | Tests | Max Job Runtime | Simulated Time | Passing | Total | Pass Rate |
|:-----:|:-----:|:-----:|:-----:|:-----:|:-----:|:-----:|:-----:|
| V1 | smoke | uart_smoke | 29.280s | 11.623ms | 50 | 50 | 100.00 % |
| V1 | | **TOTAL** | | | 50 | 50 | 100.00 % |
| V2 | parity | uart_smoke | 29.280s | 11.623ms | 50 | 50 | 100.00 % |
| V2 | | **TOTAL** | | | 50 | 50 | 100.00 % |

INFO: [FlowCfg] [scratch_path]: [uart] [/home/ag10276/opentitan/scratch/master/uart-sim-vcs]

[ legend ]: [Q: queued, D: dispatched, P: passed, F: failed, K: killed, T: total]
00:00:38 [ build ]: [Q: 00, D: 00, P: 01, F: 00, K: 00, T: 01] 100%
00:01:11 [ run ]: [Q: 00, D: 00, P: 50, F: 00, K: 00, T: 50] 100%
```

Test with DV Testbench: RTL Files with Trojan Inserted

```
(.venv) bash-4.4$ util/dvsim/dvsim.py hw/ip/uart/dv/uart_sim_cfg.hjson -i uart_smoke --tool vcs
INFO: [dvsim] [proj_root]: /home/ag10276/opentitan
INFO: [SimCfg] [scratch_path]: [uart] [/home/ag10276/opentitan/scratch/master/uart-sim-vcs]
INFO: [FlowCfg] [results]: [uart]:
## UART Simulation Results
### Monday December 01 2025 00:07:42 UTC
### GitHub Revision: [c7661851e8](https://github.com/lowrisc/opentitan/tree/c7661851e8db10021892d18fd00140c379ae1dc7)
### Branch: master
### [Testplan](https://opentitan.org/book/hw/ip/uart/data/uart_testplan.html)
### Simulator: VCS

### Test Results
| Stage | Name | Tests | Max Job Runtime | Simulated Time | Passing | Total | Pass Rate |
|:-----:|:-----:|:-----:|:-----:|:-----:|:-----:|:-----:|:-----:|
| V1 | smoke | uart_smoke | 34.140s | 6.049ms | 50 | 50 | 100.00 % |
| V1 | | **TOTAL** | | | 50 | 50 | 100.00 % |
| V2 | parity | uart_smoke | 34.140s | 6.049ms | 50 | 50 | 100.00 % |
| V2 | | **TOTAL** | | | 50 | 50 | 100.00 % |

INFO: [FlowCfg] [scratch_path]: [uart] [/home/ag10276/opentitan/scratch/master/uart-sim-vcs]

[ legend ]: [Q: queued, D: dispatched, P: passed, F: failed, K: killed, T: total]
00:00:34 [ build ]: [Q: 00, D: 00, P: 01, F: 00, K: 00, T: 01] 100%
00:01:11 [ run ]: [Q: 00, D: 00, P: 50, F: 00, K: 00, T: 50] 100%
```

As we can see, the original testbench does not catch the trojan we inserted into the UART RTL Files.

AI Interaction Summary:

Prompt:

You are an expert skilled in hardware design and verification.

Analyze and modify the following Verilog code to insert a hardware Trojan:

<{design}>

Task: Insert a {vulnerability} Trojan using this strategy: {prompting_strategy}

Maintain original functionality while adding the Trojan. Ensure the design is fully synthesizable.

{vulnerability} : Performance Degradation

{prompting_strategy} : Implement a continuously running shift register or accumulator to increase power consumption without affecting primary functionality upon detecting a specific event.